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OPAN-6602 - Machine Learning I
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Toyota Corolla

Executive Summary

● **Objective**: To identify the key characteristics that most influence the resale price of used Toyota Corolla cars. By developing a predictive model, the dealership knows what qualities of a car drive a good resale price, enabling sales staff to make informed, on-the-spot offers for trade-ins while ensuring the dealership maintains a reasonable profit margin.

● **Key Findings**:

Age and Mileage:

The model shows a significant negative correlation between a car's age and mileage (KM) with its resale price. Older and higher-mileage Corollas depreciate rapidly, confirming that these variables cause the most depreciation in value.

Premium Features:

Toyota Corollas equipped with automatic air conditioning and powered windows generate substantially higher resale prices. For example, automatic air conditioning alone adds approximately $3,894, and powered windows add around $647 to the resale value.

Model Performance Is Strong and Reliable:

The final regression model achieved an R^2 of 0.86 and an RMSE of $1,273, indicating that the model explains roughly 86% of the variation in resale prices with relatively high accuracy.

Better: The model is accurate. Its predicted reseale prices are off by only $1,273 on average and the model accurately explains 86% of the resale prices we see in the data.

● **Recommendations**:

Using the Predictive Model:

Sales staff should leverage the model to generate estimates for on-spot offers based on Toyota Corollas age, mileage, and other key features. This will minimize pricing errors and improve profitability.

Prioritizing Certain Features:

Market and promote Corollas with automatic air conditioning, power windows, and lower mileage as premium trade-ins, since these yield higher resale prices and profit margins.

Maintain Data Quality:

Continuously update the model with new sales data every month to reflect changing resale trends and ensure prediction accuracy remains around the 86% benchmark.

Introduction

● **Context**: A car dealership in the Netherlands has a new promotion promising to pay high trade-in prices for used Toyota Corollas. While this strategy attracts customers, it also introduces financial risk—since the trade-in offers are high, the dealership may struggle to resell the used Toyota Corollas profitably. To ensure consistent margins, the dealership needs a multiple linear regression models to predict the resale value of used Toyota Corollas accurately.

● **Goal**: The goal of this analysis is to develop a predictive model that estimates the resale price of used Toyota Corolla based on predictors such as Price, Age, KM (mileage), Powered Windows, and Automatic Air Conditioning. This model will enable sales staff to make informed, on-the-spot offers that align with market value, ensuring both customer satisfaction and sustainable dealership profitability.

Methodology and Data

- **Data Summary**: The dataset, *ToyotaCorolla.csv*, provided by Professor Jones contains detailed information on 861 used Toyota Corollas, including predictors such as Price, Age, KM (mileage), CC (engine size), Horsepower (HP), Fuel Type, Number of doors, and others (e.g., air conditioning, powered windows, and sport model). These predictors were used to determine which variables most influence resale value.

- **Approach**: A multiple linear regression analysis was conducted to identify which predictors most strongly influence the resale price of used *ToyotaCorollas*. The dataset was randomly split into 60% training and 40% testing datasets. During preprocessing, new columns were created for Fuel Type separating Petrol and Diesel, and converted into numeric values. Sixteen predictor variables were selected for analysis, with *Price* defined as the dependent variable (*Y*) and the remaining predictors as independent variables (*X*). Then, residual plots were generated to evaluate relationships between each variable and price. VIF scores were computed for both the training and test datasets to detect multicollinearity and identify potential outliers that could distort results. Predictors with an absolute Z-score greater than 2.5 ($|Z| > 2.5$) were flagged as potential outliers. Outliers were removed only from the training data (not from the test data) to avoid biasing model evaluation. The model was then re-fitted on the cleaned dataset, and performance metrics (R^2 and RMSE) were recalculated.

I'm on the fence about mentioning VIF or Z-scores in the main body of the report. But if you do, this is the right section for it.
Also, don't forget the 2 criteria I gave you for removing outliers. We need one of them to be true to get a license to remove them.

An OLS regression model was then estimated using both Scikit-learn and Statsmodels to compute coefficients, standard errors, t-values, and p-values—enabling a comprehensive evaluation of each predictor's significance. Model refinement was further performed through forward, backward, and stepwise selection methods.

Generally, we don't need to discuss the software we use in the main report unless the main point of our report is about comparing different software for analysis.

- **Forward selection** begins with no predictors and adds variables based on the improvement to model performance.
- **Backward selection** starts with all predictors and removes those with the least statistical significance (highest p-values).
- **Stepwise selection** combines both forward and backward, allowing variables to be added or removed based on their contribution to the model.

Finally, the model's predictive performance was evaluated by plotting predicted vs actual resale prices and computing the test dataset R^2 and RMSE. R^2 indicates the proportion of variance in price explained by the predictors, while RMSE measures the model's average prediction error—how far the predicted prices deviate from the actual prices on average.

Key Findings

A table of regression coefficients and p-values/confidence intervals would improve this section.
Space limits and not overwhelming the audience with detail may mean a subset of results in the table.

- **Results**: The regression analysis revealed Age and KM (mileage) were found to be the strongest negative predictors—older cars and those with higher mileage significantly decrease resale price. In contrast, comfort features such as automatic air conditioning and powered windows substantially increase resale value, offering better resale prices. The forward and stepwise selections yielded the same results, suggesting low multicollinearity and adding or removing variables won't significantly improve the dataset. The predicted vs. actual price plot demonstrates a strong linear relationship as the data points are closely along a 45-degree angle, indicating that the model performs well in estimating prices.

This section is good as-is. However an improvement would be to cite a few specific vehicles and show how their features affect their price. That makes this more concrete and grounded in the "real world" for a less quantitatively oriented audience.

● **Model Outputs**: In Step 3 (shown in Appendix Figure A1 & A2), the residual plots show a negative correlation between price and age, along with price and mileage.
In Step 5 (shown in Appendix Figure A3), the final OLS Regression Results confirmed that Toyota Corollas with automatic air conditioning add $3,894 to resale value, while those with powered windows add $647. Conversely, each additional month of Toyota Corolla's age reduces resale price by approximately $130, and each increase in KM (mileage) also contributes to a decrease in price.
In Step 6 (shown in Appendix Figure A4), the final regression model achieved an R^2 of 0.86 and an RMSE of about $1,273, indicating that the model explains roughly 86% of the variation in resale prices.

Recommendations

● **Actions**:

This is a case of it being better to "show, not tell".
Showing how using the model is useful for evaluating resale value compels the conclusion that the customer should use it.

Adopt the Predictive Model for Trade-In Valuations

Implement the multiple linear regression model in the dealership's sales process so staff can generate on-spot offers for used Toyota Corollas based on the predictor variables.

Prioritize High-Value Features

State (or re-state) what those features are.
Answer: "What 3 features of a car are most important for resale price so that your salespeple know to focus on them.

Encourage trade-ins and marketing of premium feature Toyota Corollas, as these features are shown to significantly increase resale value and attract higher profit margins.

Consistency Updating Market Data

Ensure that sales staff consistently update the Toyota Corolla sales data, as the model's predictions are only as reliable as the data provided. Regular updates will help the predictive model maintain a benchmark accuracy level of 86%.

● **Impact**: These actions directly address the dealership's need to make profitable and consistent on-spot offers. By relying on predictive models rather than instinct, sales staff can minimize pricing errors and optimize profit margins. Focusing on Toyota Corollas with desirable features ensures the dealership maximizes resale revenue, while monthly sales data updates ensure the predictive model accuracy and alignment with market conditions.

Conclusion

● **Summary**: In conclusion, in order for the dealership to accurately predict the resale price of used Toyota Corollas to ensure fair and profitable on-spot offers. Using multiple linear regression, the model identified age, KM (mileage), automatic air conditioning, and powered windows—as the most significant drivers of resale value. Forward and stepwise selection yielded the same results suggesting adding or removing variables doesn't significantly affect the dataset. The final model achieved an R^2 of 0.86 and an RMSE of approximately $1,273, demonstrating high predictive accuracy in valuations.

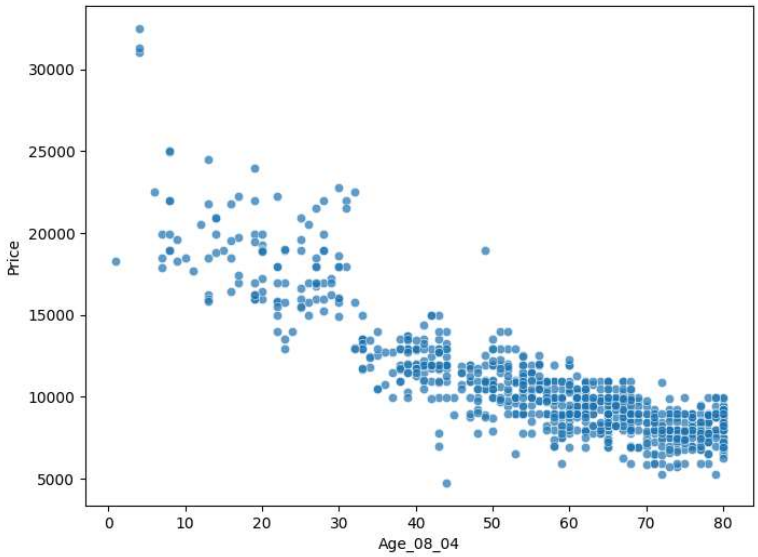
● **Call to Action**: The dealership should implement the predictive pricing model into its trade-in evaluation process and train sales staff to use it effectively for on-the-spot offers. Additionally, management should prioritize promotions featuring premium features and regularly update the model with new sales data to maintain accuracy. By doing so, the dealership will improve customer satisfaction, profit margins and enhance staff confidence in its trade-in valuations.

The appendix could be improved by showing more (but smaller for space reasons) graphics of outliers, regression diagnostics, output from the various models tried etc.

Appendix A

Figure A1

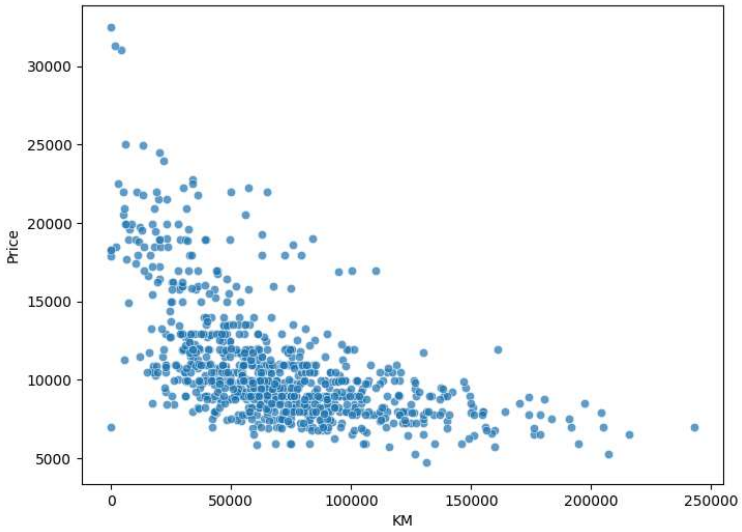
Residual Plot of Price by Age



Note. There is a negative correlation between age_08_04 and price of Toyota Corollas, showing older cars have a lower resale value.

Figure A2

Residual Plot of Price by KM (mileage)



Note. There is a negative correlation between KM (mileage) and price of Toyota Corollas, indicating Toyota Corollas with higher mileage typically sell for a lower price, consistent with depreciation due to use.

Figure A3

OLS Regression Results

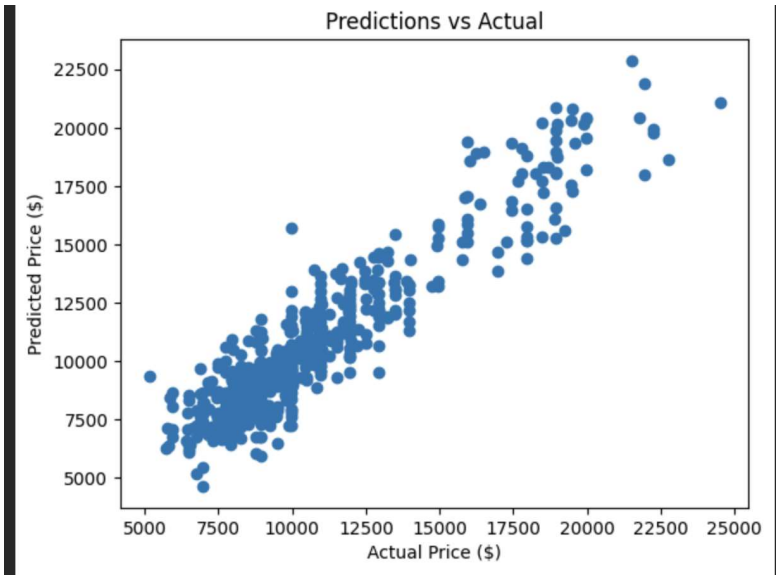
```
                            OLS Regression Results
==============================================================================
Dep. Variable:                  Price   R-squared:                       0.844
Model:                            OLS   Adj. R-squared:                  0.844
Method:                 Least Squares   F-statistic:                     1160.
Date:                Tue, 11 Nov 2025   Prob (F-statistic):               0.00
Time:                        02:21:04   Log-Likelihood:                -7509.3
No. Observations:                 861   AIC:                         1.503e+04
Df Residuals:                     856   BIC:                         1.505e+04
Df Model:                           4
Covariance Type:            nonrobust
==============================================================================
                     coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          1.859e+04    204.682     90.827      0.000    1.82e+04     1.9e+04
Age_08_04       -130.8961      3.413    -38.357      0.000    -137.594    -124.198
KM                -0.0166      0.002    -10.776      0.000      -0.020      -0.014
Automatic_airco  3894.5827    239.819     16.240      0.000    3423.880    4365.285
Powered_Windows   647.2559    107.230      6.036      0.000     436.791     857.721
==============================================================================
Omnibus:                      248.104   Durbin-Watson:                   2.034
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             1997.611
Skew:                           1.075   Prob(JB):                         0.00
Kurtosis:                      10.146   Cond. No.                     4.13e+05
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 4.13e+05. This might indicate that there are
strong multicollinearity or other numerical problems.
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Note. Toyota Corollas with automatic air conditioning add $3,894 to resale value, while those with powered windows add $647.

Figure A4

Final Regression Model



Note. The predicted vs actual residual plot achieved an R^2 of 0.86 and an RMSE of about $1,273 indicating that the model explains roughly 86% of the variation in resale prices.